



Evaluate $\int x^2 \ln x \, dx.$

- A** $\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$
- B** $\frac{x^3}{3} \ln x - \frac{x^3}{3} + C$
- C** $\frac{x^3}{3} \ln x + C$
- D** $x^3 \ln x - \frac{x^3}{9} + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$

Between the two factors, log wins the LIATE priority — differentiate $\ln x$ (it simplifies), integrate the polynomial.

STEP BY STEP

1. Let $u = \ln x$, $dv = x^2 dx \Rightarrow du = \frac{dx}{x}$, $v = \frac{x^3}{3}$.
2. Apply IBP: $\frac{x^3}{3} \ln x - \int \frac{x^3}{3} \cdot \frac{1}{x} dx$.
3. Simplify and integrate: $\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$.





Evaluate $\int e^x \cos x \, dx.$

- A** $\frac{e^x(\sin x + \cos x)}{2} + C$
- B** $e^x(\sin x + \cos x) + C$
- C** $\frac{e^x(\sin x - \cos x)}{2} + C$
- D** $e^x \sin x + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $\frac{e^x(\sin x + \cos x)}{2} + C$

Neither e^x nor $\cos x$ ever disappears under repeated by-parts — the trick is applying it twice and solving for the original integral algebraically.

STEP BY STEP

1. First pass: $u = e^x, dv = \cos x dx \Rightarrow \int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx.$
2. Second pass on the new integral: $\int e^x \sin x dx = -e^x \cos x + \int e^x \cos x dx.$
3. Substitute back: $\int e^x \cos x dx = e^x \sin x + e^x \cos x - \int e^x \cos x dx.$
4. Solve algebraically: $2 \int e^x \cos x dx = e^x(\sin x + \cos x),$ so the integral is $\frac{e^x(\sin x + \cos x)}{2} + C.$





Using the reduction formula $I_n = x^n e^x - nI_{n-1}$,
evaluate $\int x^3 e^x dx$.

- A** $e^x(x^3 - 3x^2 + 6x - 6) + C$
- B** $e^x(x^3 + 3x^2 + 6x + 6) + C$
- C** $e^x(x^3 - 3x^2 + 6x) + C$
- D** $e^x(x^2 - 2x + 2) + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $e^x(x^3 - 3x^2 + 6x - 6) + C$

Rather than integrating by parts three separate times, the reduction formula lets you climb from I_0 up to I_3 mechanically.

STEP BY STEP

1. Base case: $I_0 = \int e^x dx = e^x$.
2. $I_1 = xe^x - 1 \cdot I_0 = xe^x - e^x$.
3. $I_2 = x^2e^x - 2 \cdot I_1 = x^2e^x - 2xe^x + 2e^x$.
4. $I_3 = x^3e^x - 3 \cdot I_2 = x^3e^x - 3x^2e^x + 6xe^x - 6e^x$.
5. Factor: $e^x(x^3 - 3x^2 + 6x - 6) + C$.





Using the reduction formula

$$I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}, \text{ evaluate}$$
$$\int \sin^4 x \, dx.$$

- A** $-\frac{\sin^3 x \cos x}{4} - \frac{3}{8} \sin x \cos x + \frac{3x}{8} + C$
- B** $-\frac{\sin^3 x \cos x}{4} + \frac{3}{8} \sin x \cos x + \frac{3x}{8} + C$
- C** $-\frac{\sin^3 x \cos x}{4} - \frac{3}{8} \sin x \cos x + C$
- D** $\frac{\sin^3 x \cos x}{4} - \frac{3}{8} \sin x \cos x + \frac{3x}{8} + C$



SWIPE FOR THE ANSWER →



✓ **ANSWER: (A)** $-\frac{\sin^3 x \cos x}{4} - \frac{3}{8} \sin x \cos x + \frac{3x}{8} + C$

The power-reduction formula for $\sin^n x$ drops the exponent by 2 each round, bottoming out at the easy base case $I_0 = x$.

STEP BY STEP

1. Base case: $I_0 = \int dx = x$.
2. $I_2 = -\frac{\sin x \cos x}{2} + \frac{1}{2}I_0 = -\frac{\sin x \cos x}{2} + \frac{x}{2}$.
3. $I_4 = -\frac{\sin^3 x \cos x}{4} + \frac{3}{4}I_2$.
4. Substitute I_2 : $I_4 = -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \left(-\frac{\sin x \cos x}{2} + \frac{x}{2} \right)$.
5. Simplify: $I_4 = -\frac{\sin^3 x \cos x}{4} - \frac{3}{8} \sin x \cos x + \frac{3x}{8} + C$.





Evaluate $\int \frac{3x + 5}{(x - 1)(x + 2)} dx$ (distinct linear factors)

- A** $\frac{8}{3} \ln |x - 1| + \frac{1}{3} \ln |x + 2| + C$
- B** $\frac{1}{3} \ln |x - 1| + \frac{8}{3} \ln |x + 2| + C$
- C** $\frac{8}{3} \ln |x - 1| - \frac{1}{3} \ln |x + 2| + C$
- D** $8 \ln |x - 1| + \frac{1}{3} \ln |x + 2| + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $\frac{8}{3} \ln |x - 1| + \frac{1}{3} \ln |x + 2| + C$

Two distinct linear factors mean two simple fractions — cover-up substitution (plugging in the root of each factor) finds A and B instantly.

STEP BY STEP

1. Set up the decomposition: $\frac{3x + 5}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$.
2. Clear denominators: $3x + 5 = A(x + 2) + B(x - 1)$.
3. Let $x = 1$ (kills B): $8 = 3A \Rightarrow A = \frac{8}{3}$.
4. Let $x = -2$ (kills A): $-1 = -3B \Rightarrow B = \frac{1}{3}$.
5. Split integral: $\int \left[\frac{8/3}{x - 1} + \frac{1/3}{x + 2} \right] dx$.
6. Integrate each term: $\frac{8}{3} \ln |x - 1| + \frac{1}{3} \ln |x + 2| + C$.





Evaluate $\int \frac{3x + 5}{(x - 1)^2} dx$ (repeated linear factor)

- A $3 \ln |x - 1| - \frac{8}{x - 1} + C$
- B $3 \ln |x - 1| + \frac{8}{x - 1} + C$
- C $8 \ln |x - 1| - \frac{3}{x - 1} + C$
- D $3 \ln |x - 1| - \frac{8}{(x - 1)^2} + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $3 \ln |x - 1| - \frac{8}{x - 1} + C$

A repeated factor needs one term for every power up to the repeat — matching coefficients (not just cover-up alone) finds both constants.

STEP BY STEP

1. Set up the decomposition: $\frac{3x + 5}{(x - 1)^2} = \frac{A}{x - 1} + \frac{B}{(x - 1)^2}$.
2. Clear denominators: $3x + 5 = A(x - 1) + B$.
3. Let $x = 1$ (kills A): $8 = B$.
4. Match the x -coefficient: $A(x - 1) + B = Ax + (B - A)$, so $A = 3$.
5. Split integral: $\int \left[\frac{3}{x - 1} + \frac{8}{(x - 1)^2} \right] dx$.
6. Integrate each term: $3 \ln |x - 1| - \frac{8}{x - 1} + C$.





Evaluate $\int \frac{x^2 + x + 1}{x(x^2 + 1)} dx$ (irreducible quadratic factor)

- A** $\ln |x| + \arctan x + C$
- B** $\ln |x| - \arctan x + C$
- C** $\ln |x| + \ln(x^2 + 1) + C$
- D** $\arctan x + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $\ln |x| + \arctan x + C$

An irreducible quadratic factor needs a linear numerator $(Bx + C)$, not just a constant — that's exactly what produces the arctan term.

STEP BY STEP

1. Set up the decomposition: $\frac{x^2 + x + 1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$.
2. Clear denominators: $x^2 + x + 1 = A(x^2 + 1) + (Bx + C)x$.
3. Let $x = 0$ (kills B, C): $1 = A$.
4. Match x^2 coefficients: $1 = A + B \Rightarrow B = 0$.
5. Match x coefficients: $1 = C$.
6. Integrate: $\int \left[\frac{1}{x} + \frac{1}{x^2 + 1} \right] dx = \ln |x| + \arctan x + C$.





Evaluate $\int 2xe^{x^2} dx$.

- A** $e^{x^2} + C$
- B** $2xe^{x^2} + C$
- C** $x^2e^{x^2} + C$
- D** $\frac{e^{x^2}}{2} + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $e^{x^2} + C$

The derivative of the exponent, $2x$, is sitting right there in the integrand — a strong signal to substitute $u = x^2$.

STEP BY STEP

1. Let $u = x^2 \Rightarrow du = 2x dx$.
2. Substitute: $\int e^u du = e^u$.
3. Back-substitute: $e^{x^2} + C$.





Evaluate $\int \frac{\ln x}{x} dx.$

- A** $\frac{(\ln x)^2}{2} + C$
- B** $\ln x + C$
- C** $(\ln x)^2 + C$
- D** $\frac{1}{x} + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $\frac{(\ln x)^2}{2} + C$

$1/x$ is exactly the derivative of $\ln x$ — once you see that pairing, the substitution writes itself.

STEP BY STEP

1. Let $u = \ln x \Rightarrow du = \frac{dx}{x}$.
2. Substitute: $\int u du = \frac{u^2}{2}$.
3. Back-substitute: $\frac{(\ln x)^2}{2} + C$.





Evaluate $\int \sin^3 x \cos x \, dx$.

- A** $\frac{\sin^4 x}{4} + C$
- B** $\sin^4 x + C$
- C** $-\frac{\sin^4 x}{4} + C$
- D** $\frac{\cos^4 x}{4} + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $\frac{\sin^4 x}{4} + C$

One factor of $\cos x$ is exactly $d(\sin x)$ — peel it off as du and the rest becomes a simple power rule in u .

STEP BY STEP

1. Let $u = \sin x \Rightarrow du = \cos x dx$.
2. Substitute: $\int u^3 du = \frac{u^4}{4}$.
3. Back-substitute: $\frac{\sin^4 x}{4} + C$.





Evaluate $\int \frac{x}{x^2 + 1} dx.$

- A** $\frac{1}{2} \ln(x^2 + 1) + C$
- B** $\ln(x^2 + 1) + C$
- C** $\ln |x| + C$
- D** $\frac{1}{2} \ln |x| + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $\frac{1}{2} \ln(x^2 + 1) + C$

The numerator x is, up to a constant, the derivative of the denominator — a classic cue to substitute $u = x^2 + 1$.

STEP BY STEP

1. Let $u = x^2 + 1 \Rightarrow du = 2x dx$, so $x dx = \frac{du}{2}$.
2. Substitute: $\int \frac{1}{u} \cdot \frac{du}{2} = \frac{1}{2} \ln |u|$.
3. Back-substitute: $\frac{1}{2} \ln(x^2 + 1) + C$.





Evaluate $\int e^{\tan x} \sec^2 x \, dx.$

- A** $e^{\tan x} + C$
- B** $e^{\tan x} \sec^2 x + C$
- C** $\frac{e^{\tan x}}{\sec^2 x} + C$
- D** $\tan x \cdot e^{\tan x} + C$

SWIPE FOR THE ANSWER →





✓ **ANSWER: (A)** $e^{\tan x} + C$

$\sec^2 x$ is exactly the derivative of $\tan x$, so the whole exponent-and-derivative pair substitutes away in one clean move.

STEP BY STEP

1. Let $u = \tan x \Rightarrow du = \sec^2 x dx$.
2. Substitute: $\int e^u du = e^u$.
3. Back-substitute: $e^{\tan x} + C$.

